

sbuck-5v3a — Design Document

ai-ee v1 pipeline
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1 Board and Run Metadata

Field	Value
board	sbuck-5v3a
workspace	boards/sbuck-5v3a
phase	P2
gate status	no gates recorded yet
state created	2026-08-08T22:12:30
state updated	2026-08-08T23:29:56

2 Overview

Brief - synchronous buck converter board (sbuck-5v3a)

Design a synchronous buck converter board.

SPECIFICATION

- Input: 7-18V DC, nominal 12V, via 2-pin 5.08mm screw terminal
- Output: 5.0V +/-2%, 3A continuous, via 2-pin 5.08mm screw terminal
- Load step: 0 to 3A, recovery within 100us, <200mV excursion
- Output ripple: <50mV pk-pk at full load
- Efficiency: >88% at 12V in, 3A out
- Ambient: up to 50C, no forced airflow

- Protection: input reverse polarity, output overcurrent, thermal shutdown
- Board: $\leq 50 \times 40 \text{mm}$, 4x M3 mounting holes, 2 or 4 layer (your call, justify it)

PART SELECTION

- Prefer an integrated synchronous buck IC over a controller + discrete FETs at this power level. Candidate families to evaluate: TI TPS54xxx/TPS56xxx, MPS MP23xx/MP15xx, Diodes AP63xxx, Silergy SY82xx. Do not trust remembered ratings for any of these - pull the datasheet and confirm V_{in} range, I_{out} , and switching frequency before committing.
- Verify LCSC stock and basic/extended tier for every part before finalising. A part that is out of stock is a failed design.
- Inductor: compute ripple current $dI = (V_{in} - V_{out}) * D / (L * f_{sw})$, where $D = V_{out}/V_{in}$. Target 20-40% of full load current. Size saturation current above $I_{out} + dI/2$ with margin, at temperature, not at 25C. Check DCR against your efficiency target.
- Input capacitance: ceramic close-in for the switching loop plus bulk for the cable inductance. Input cap RMS current is roughly $I_{out} * \sqrt{D * (1-D)}$ - check the ceramic's ripple rating and derate MLCC capacitance for DC bias, which can cost more than half the nameplate value at rated voltage.
- Output capacitance: ripple is roughly $dI / (8 * f_{sw} * C_{out}) + dI * ESR$. Check the load-step requirement too, which usually sets C_{out} above what ripple alone needs.
- Every MLCC needs voltage derating; X7R or X5R only, no Y5V.

SCHEMATIC

- Feedback divider sized for the IC's internal reference; keep divider impedance low enough that FB bias current does not shift the setpoint.
- Include the bootstrap capacitor, enable/UVLO divider if the IC supports it, and a soft-start component if it is not internal.
- Reverse polarity: a P-channel FET in the supply path is more efficient than a series diode at 3A. Confirm V_{gs} rating against maximum input voltage.
- Add an optional RC snubber footprint across the low-side switch, DNP by default.
- Provide test points for VIN, SW, VOUT, FB, EN, GND, plus a low-inductance scope ground point near the output.

LAYOUT

Placement first, routing second. Placement decides this design.

- The high di/dt loop is input capacitor $\rightarrow$ high-side switch $\rightarrow$ low-side switch $\rightarrow$ back to the capacitor's ground. Minimise its enclosed area above everything else. Put the smallest input ceramic directly across the IC's VIN and PGND pins on the same layer, no vias in that loop.
- The switch node is the primary radiator. Make its copper large enough to carry current and conduct heat, and no larger. Do not pour it as a plane.
- Keep the feedback trace short, away from SW and the inductor body, and tap it from the output capacitor terminal rather than the inductor pin so it senses the regulated node.
- Bootstrap capacitor directly at its pins.

- Single low-impedance ground reference under the converter. Avoid routing signal returns through the power ground path. If 4 layer, put an uninterrupted ground plane on the layer directly under the switching components and do not cut it.
- Size copper by current using IPC-2152, accounting for the 10-20C rise you can accept. 1oz outer copper is 35um; check whether inner layers are 0.5oz in the stackup you pick.
- Place a via array under the IC's thermal pad and stitch the ground planes together around the loop.

THERMAL

- Estimate dissipation at 12V/3A from the IC's efficiency curve, then check junction temperature against R_{thJA} for the copper area you actually have, at 50C ambient.
- Inductor DCR loss is $I^2 * DCR$; confirm the part's own temperature rise.
- If the numbers do not close, increase copper area or change parts. Do not assume the exposed pad alone is sufficient.

EMI / DFM

- Keep the switching loop and SW node away from the board edge and connectors.
- No unnecessary ground plane splits or slots under the converter.
- Silkscreen legible, none on pads, polarity and pin 1 marked on every polarised part.
- Respect JLCPCB's minimum trace/space, annular ring, and hole sizes for the process class you select. Confirm every footprint's rotation against JLCPCB's CPL convention, not the KiCad default.

BEFORE DECLARING DONE

1. DRC and ERC clean, zero unrouted nets. 2. Recompute inductor peak current, input cap RMS current, output ripple, and junction temperature from the parts you actually chose, and state the numbers. 3. Confirm each part is in stock at LCSC with its part number recorded. 4. Report the input loop enclosed area, switch node copper area, feedback trace length, and thermal via count.

DECISION POLICY

Make every remaining decision yourself. Do not ask clarifying questions. Where this brief is silent, take the conventional, conservative option. Record each non-obvious choice in DECISIONS.md with one line of reasoning.

Deliver: schematic, layout, gerbers, BOM, CPL, DECISIONS.md, and the computed numbers above.

3 Requirements

requirements.md - sbuck-5v3a

Source: `brief/brief.md` (sole input; no attached datasheets, drawings or reference schematics were supplied).

Note on process: the brief carries an explicit DECISION POLICY - "Make every remaining decision yourself. Do not ask clarifying questions. Where this brief is silent, take the conventional, conservative option." Section 9 below therefore lists the unknowns as questions WITH a

recommended default each, so the orchestrator can rule on them as delegate and record them in DECISIONS.md. Nothing in section 9 is blocking on the user.

Legend: **STATED** = written in the brief. **DERIVED** = arithmetic from stated numbers. **GUESS / ASSUMED** = not in the brief, low risk, recorded so it can be overridden. **HARD** = binds permanently at P5 board_init.

1. Function

A standalone, open-frame step-down DC-DC power module: it takes an unregulated 7-18 V DC input (12 V nominal) on a 2-pin screw terminal and produces a regulated 5.0 V $\pm 2\%$ output at up to 3 A continuous (15 W) on a second 2-pin screw terminal, using a synchronous buck topology built around an integrated buck IC (internal high-side and low-side switches) rather than a controller plus discrete FETs. The board protects itself and its source against reverse-polarity connection of the input, and protects itself against output overcurrent and over-temperature. There is no MCU, no digital interface, no telemetry, and no sequencing or remote control: the board regulates whenever a valid input is applied. Scope is the converter only.

2. Interfaces

#	Interface	Dir	Electrical	Connector
I1	DC input	in	7-18 V DC, 12 V nominal. Worst-case input current ~2.6 A at 7 V in / 3 A out (DERIVED). Reverse-polarity tolerant.	2-pin 5.08 mm pitch screw terminal (STATED). Exact part/entry direction: Q9
I2	DC output	out	5.0 V $\pm 2\%$ (4.90-5.10 V), 3 A continuous.	2-pin 5.08 mm pitch screw terminal (STATED)
I3	Test point VIN	probe	post-reverse-polarity-FET input rail	test pad, see Q13
I4	Test point SW	probe	switch node, high dv/dt - short-stub tap only, must not extend SW copper	test pad, see Q13
I5	Test point VOUT	probe	regulated output, tapped at the output cap terminal	test pad, see Q13
I6	Test point FB	probe	feedback node - high impedance, probe loading must not shift setpoint	test pad, see Q13
I7	Test point EN	probe	enable / UVLO node	test pad, see Q13
I8	Test point GND	probe	general ground reference	test pad, see Q13
I9	Scope ground	probe	dedicated LOW-INDUCTANCE ground point adjacent to the VOUT test point, sized for a scope spring/bayonet ground rather than a ground lead (STATED)	see Q13

Non-connector interface items stated in the brief:

- RC snubber footprint across the low-side switch (i.e. SW to PGND), populated DNP by default. Footprint must exist and be routable with low inductance even though unstuffed.
- No indicator LED, no PG output, no external enable input, no adjustable-output feature, and no remote sense are stated. See Q11 (LED) and Q14 (EN control) for the defaults.

3. Power

Input (STATED)

Param	Value
Range	7.0 - 18.0 V DC
Nominal	12.0 V
Absolute max at connector	18 V (no surge/load-dump spec given - see Q6)

| Source impedance / cabling | not stated; brief requires bulk input capacitance for "the cable inductance", so a non-trivial cable run is implied |

Output (STATED)

Param	Value
Voltage	5.0 V, +/-2% = 4.90-5.10 V (window scope: see Q1)
Current	3.0 A continuous
Power	15.0 W (DERIVED)
Ripple	< 50 mV pk-pk at full load (measurement bandwidth: see Q2)
Load step	0 -> 3 A and 3 A -> 0: excursion < 200 mV, recovery within 100 us (slew rate of the step: see Q3)
Efficiency	> 88% at 12 V in, 3 A out (measurement points: see Q4)

Derived operating points (arithmetic from the stated spec; not guesses)

Param	At Vin = 7 V	At Vin = 12 V	At Vin = 18 V
Duty D = Vout/Vin (ideal)	0.714	0.417	0.278
Input current at 3 A out, 88% eff	2.44 A	1.42 A	0.95 A
Input cap RMS current, $I_{out} \cdot \sqrt{D(1-D)}$	1.36 A	1.48 A	1.34 A

- Worst-case input capacitor RMS current over the whole input range is 1.50 A, at $D = 0.5$, i.e. $V_{in} = 10$ V - which is INSIDE the operating range. Size the input ceramics for 1.5 A RMS, not for the 1.48 A at the 12 V nominal point.
- Board dissipation at the efficiency floor: $P_{in} = 15.0/0.88 = 17.05$ W, so $P_{loss} = 2.05$ W total at 12 V/3 A. This is the thermal budget the 50 C ambient case must close against.
- Reverse-polarity P-FET conducts the full input current (2.44 A worst case at 7 V) and its $I^2 R_{ds(on)}$ loss counts against the 88% number if efficiency is measured terminal to terminal (Q4).

Rails Single output rail. No auxiliary rails, no bias rail, no battery, no charging, no energy storage beyond the converter's own input/output capacitance.

Protection (all STATED as requirements)

P	Requirement	Notes
P1	Input reverse polarity	Brief directs a P-channel FET in the supply path (not a series diode) and requires its Vgs rating be confirmed against max Vin. Vgs clamp/divider needed if Vgs rating < 18 V.
P2	Output overcurrent	Mechanism not specified - see Q7
P3	Thermal shutdown	Not specified whether IC-internal is sufficient - see Q8

4. Environment

Param	Value	Status
Max operating ambient	50 C	STATED
Airflow	none - natural convection only	STATED
Min operating ambient	not stated	Q15, default 0 C
Storage range	not stated	ASSUMED: -40 to +85 C, standard commercial part ratings
Enclosure	not stated	Q16 - drives whether 50 C is ambient-air or inside-a-box

Ingress protection	not stated	ASSUMED: none (open frame, indoor, dry)	
Vibration / shock	not stated	ASSUMED: none beyond normal handling; no staking or conformal coat	
Altitude	not stated	ASSUMED: <= 2000 m, no creepage derating	

Note: the thermal requirement is the one that bites. 2.05 W into natural convection at 50 C ambient on a board of at most 50x40 mm, with no heatsink, is the binding constraint of this design, and the brief explicitly forbids assuming the exposed pad alone closes it.

5. Size and mounting

Param	Value	Status	
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Max outline	50.0 x 40.0 mm	**HARD** (STATED as "<=50x40mm"). Binds at P5 board.init. Smaller is permitted but see Q17 - recommended default is to use the full area for thermal copper.	
Layer count	2 or 4, designer's call, must be justified	STATED as a delegated decision - see Q18	
Mounting holes	4x M3	**HARD** on count and thread size (STATED). Positions, plating and keepout: Q19	
Max component height	not stated	Q20, default 15 mm	
Board thickness	not stated	ASSUMED: 1.6 mm FR4, JLCPCB standard	
Copper weight	not stated	Q21 - 1 oz outer default, 2 oz escalation path if thermals do not close	
Keepouts / connector clearance	not stated	ASSUMED: screw terminals at board edges with wire entry clear of the mounting hardware	

6. Quantity and budget

Param	Value	Status	
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Build quantity	not stated	Q22, default 5 (JLCPCB minimum prototype run)	
Target unit cost	not stated	Q23, default: no hard cap; keep BOM under ~\$12/board at qty 5 and flag anything that would push a single line item over ~\$3	
Lifetime / production intent	not stated	ASSUMED: prototype / low-volume bench module, not a mass-production cost-down exercise	

Hard commercial constraint that IS stated: every part must be in stock at LCSC with its part number recorded; an out-of-stock part is a failed design. Stock threshold: Q24.

7. Assembly

Param	Value	Status	
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Assembly method	not stated	Q25, default: JLCPCB PCBA for all SMT	
Sides assembled	not stated	Q26, default: single-sided (top) SMT assembly	
Through-hole parts	the two 5.08 mm screw terminals are inherently THT	Q25 covers whether to buy JLC's THT service or hand-solder the two connectors	
Footprint rotation	must be checked against JLCPCB's CPL convention, NOT the KiCad default	STATED	
Process class	must respect JLCPCB minimum trace/space, annular ring and hole size for the selected class	STATED; class choice: Q27	
Part tier	LCSC Basic vs Extended not stated	Q28, default: prefer Basic/Preferred for passives, accept Extended where the function demands it (buck IC, inductor, P-FET)	

8. Compliance / safety flags

Applicable:

- **High current (≥ 3 A).** The output is specified at exactly 3.0 A continuous, which sits on the brief's own high-current threshold, and the input side draws up to ~ 2.6 A at low line. This is the flag that drives copper sizing (IPC-2152, with the 10-20 C rise the brief permits), connector current rating, and the fusing question. Per role rules this is NOT guessed: see Q5 (is 3 A the true worst case, or is there a peak/inrush above it), Q7 (overcurrent mechanism and whether an input fuse is required) and Q29 (fault behaviour the board must survive: sustained short circuit on the output).
- **Undefined fault energy.** The brief does not state whether the input source is current limited, so the power available into a fault is unbounded as specified: a 4 A input fuse at 18 V still passes ~ 72 W. Behaviour under a bolted output short, and under a reversed or swapped input, must be defined rather than assumed - Q7, Q29, Q30.
- **Switching EMI.** A hard-switching buck is a deliberate broadband noise source. The brief imposes layout-level EMI requirements (loop area, SW node away from edges/connectors, no plane splits) but names no standard - Q31.

Not applicable (confirmed against the brief):

- No mains voltage anywhere on the board. Maximum potential is 18 V.
- No battery of any chemistry, no charging circuit, no cell balancing.
- No motors, relays, solenoids or other inductive actuator loads driven by this board.
- No node above 30 V under normal operation (18 V max input; switch-node ringing must still be kept below the chosen IC's absolute max, which is a design check, not a safety-agency flag).
- No intentional RF transmission.

Consequence: no agency-certification path (UL/IEC mains safety, battery transport, RF type approval) applies to this board. The safety work is entirely thermal, fault-current and connector-rating engineering.

9. Open questions

Each has a recommended default. The orchestrator answers all of them at once as delegate, per the brief's DECISION POLICY, and each answer becomes a DECISIONS.md line.

1. ****Does the $\pm 2\%$ output window include ripple and load-step transients, or only DC accuracy?**** Default: $\pm 2\%$ is the DC setpoint over line, load and temperature; the 50 mV ripple and the 200 mV load-step excursion are separate, additional allowances. 2. **How is the <50 mV ripple measured?** Default: at the output screw terminal, scope in 20 MHz bandwidth-limited mode with a short spring ground (the standard method) - so high-frequency switching spikes above 20 MHz are excluded from the number. 3. **How fast is the 0-3 A load step?** Default: 2.5 A/us electronic-load slew (a common bench setting), applied at 12 V in. A faster step would be dominated by output-cap ESL rather than by the control loop. 4. ****Is the $>88\%$ efficiency measured at the screw terminals (including the reverse-polarity FET and terminal resistance) or across the IC only?**** Default: terminal to terminal, i.e. the reverse-polarity FET loss counts against the budget. This is the conservative reading. 5. ****Is 3 A the true worst case, or must the board survive brief peaks above it (e.g. a capacitive or lamp load at turn-on)?**** Default: 3.0 A is the continuous maximum and the design target; the IC's own current limit is allowed to be the only headroom above it, with the current limit set no lower than 4 A to avoid nuisance foldback. 6. **Any input**

surge above 18 V (automotive load dump, hot-plug ringing on a long cable)? Default: no formal surge requirement, but select the IC and input caps for a minimum 28 V rating ($>1.5\times$ the 18 V max) so hot-plug ringing on an inductive cable does not destroy the part. 7. **Overcurrent protection mechanism, and is an input fuse required?** Default: rely on the buck IC's internal cycle-by-cycle current limit with hiccup-mode short-circuit protection (auto-recovering), AND fit a slow-blow input fuse or PTC sized ~ 4 A as a fail-safe against a shorted high-side switch, which the IC cannot protect against. 8. ****Is the IC's internal thermal shutdown sufficient, or is an independent thermal cutout required?**** Default: internal TSD is sufficient given the low power and the absence of any agency requirement; no separate thermistor or cutout. 9. **Screw terminal part and wire entry direction.** Default: 5.08 mm pitch, 2-pin, horizontal (side) wire entry facing outward from the board edge, rated ≥ 10 A / 300 V, accepting 12-24 AWG; choose whichever LCSC-stocked family (KF128/KF301-class or equivalent) is in stock in quantity with a verified footprint. 10. **Input and output on opposite board edges, or the same edge?** Default: opposite short edges (input left, output right), giving a straight-through power flow, maximum separation between the noisy input loop and the output terminal, and no chance of a user swapping the two identical connectors... which leads to Q30. 11. **Any indicator LED?** Default: yes, one low-current green LED on VOUT (~ 1 mA, ~ 5 mW), as an at-a-glance "output is live" indicator. Cost to efficiency at 3 A is negligible (0.03%). Say no if the pure-efficiency reading is preferred. 12. **Any power-good (PG) output brought out?** Default: no external PG connector; if the chosen IC has a PG pin, bring it to a test pad only. 13. **Test point physical form.** Default: 1.5 mm bare round SMD pads for VIN, SW, FB, EN and GND; the VOUT test point paired with a dedicated scope-ground pad at ~ 5 mm spacing so a scope spring ground reaches both (this satisfies the brief's "low-inductance scope ground point near the output"). No THT loop test points - they add inductance and height and cost a hand-solder step. 14. **Is EN to be user-controllable (jumper/header), or purely an internal UVLO node?** Default: internal only - a resistor divider from VIN sets UVLO at approximately 6.5 V rising / 6.0 V falling, so the converter refuses to start below the 7 V spec floor, and EN is exposed as a test pad only. 15. **Minimum operating ambient?** Default: 0 C operating, with all parts rated -40 C or lower so the limit is the specification and not the silicon. 16. **Enclosure: is the 50 C figure free-air ambient or inside a sealed box?** Default: open-frame, free-air 50 C, mounted on standoffs with unobstructed convection on both faces. If it is ever boxed, the internal air temperature must be re-stated - a sealed enclosure would likely break the thermal budget. 17. **May the board use the full 50 x 40 mm, or should it be as small as possible?** Default: use the full 50 x 40 mm envelope. Copper area is the free variable that closes the thermal case, and the brief caps the size rather than asking for minimum size. 18. **2-layer or 4-layer?** Default: 4-layer. The brief's own layout section demands "an uninterrupted ground plane on the layer directly under the switching components", a single low-impedance ground reference, and enough copper to sink ~ 2 W by convection - all three are materially easier and lower-risk on 4 layers, and the JLCPCB cost delta at prototype quantity is small. Justification to be recorded at P5. 19. **Mounting hole geometry.** Default: 3.2 mm holes, non-plated (NPTH), one at each corner inset 3.5 mm from both edges, 6.5 mm diameter copper/mask keepout for an M3 washer, isolated from GND (no chassis-ground bond, avoiding an uncontrolled ground loop through the standoffs). 20. **Maximum component height?** Default: 15 mm above the top surface, which comfortably fits a 5.08 mm screw terminal (~ 10 mm) and any shielded power inductor likely to be chosen (< 7 mm). 21. **Copper weight?** Default: JLCPCB standard stackup - 1 oz (35 μ m) outer, 0.5 oz inner on 4-layer (the brief explicitly warns to check this). Escalate to 2 oz outer only if the IPC-2152 sizing or the junction-temperature calculation fails at 1 oz; record the escalation if it happens. 22. **Build quantity?** Default: 5 boards (JLCPCB prototype minimum). 23. **Target unit cost?** Default: no hard cap; keep the BOM under roughly \$12/board at qty 5 and flag any single line item over $\sim$ \$3 for a conscious decision. 24. **LCSC stock threshold to call a part "available"?** Default: ≥ 500 pcs in stock for the buck IC, inductor and P-FET; ≥ 2000 pcs for passives; and no part whose LCSC page shows an end-of-life or "last stock" state. 25. **Assembly method, including the two THT screw terminals?** Default: JLCPCB PCBA for all SMT parts; the two screw terminals hand-soldered on receipt (avoids JLC's THT surcharge on a 5-board run and

keeps the connector choice free of JLC's THT part list). Both connector footprints must still be documented in the BOM/CPL as DNP-for-assembly. 26. **Single- or double-sided assembly?** Default: single-sided, top only. All active and passive parts on top; bottom reserved for ground plane, thermal copper and the test pads' ground stitching. This halves assembly cost and keeps the bottom face free to conduct heat. 27. **JLCPCB process class?** Default: the standard/economic class (typically 5 mil (0.127 mm) minimum trace and space, 0.3 mm minimum drill, 0.13 mm annular ring). Nothing in this design needs finer, and staying standard keeps cost and yield where they should be. Actual limits to be confirmed against JLC's published capability table at P5, not from memory. 28. **LCSC Basic vs Extended parts?** Default: prefer Basic/Preferred for R, C, L and the LED; accept Extended for the buck IC, the power inductor and the P-channel FET where the requirement demands a specific part. Extended parts add a per-line feeder fee - acceptable at these quantities. 29. **Required behaviour under a sustained output short circuit?** Default: the board must survive indefinitely and recover automatically when the short is removed (hiccup-mode current limit), without exceeding component ratings and without the input fuse blowing unless a genuine hard failure has occurred. 30. ****Required behaviour if the OUTPUT terminal is back-driven, or if input and output connectors are swapped by the user (they are identical 2-pin 5.08 mm parts)?**** Default: no active output reverse-current or back-drive protection is required, but the silkscreen must unambiguously mark IN/OUT and polarity on both connectors, and the board should not be destroyed by a brief 5 V back-feed into the output. A swapped connection driving 12 V into the 5 V output is NOT required to be survivable - it will be prevented by labelling. Flag if survivability is actually wanted. 31. **Any EMI standard to meet (CISPR 32 / EN 55032 class B conducted or radiated)?** Default: no formal standard and no compliance testing; apply the brief's layout rules as best practice, leave the DNP snubber footprint as the mitigation hook, and add no input EMI filter beyond the bulk plus ceramic input capacitance already required. 32. **Switching frequency preference?** Default: no hard requirement - choose whatever the selected IC family supports in the roughly 400-800 kHz range, trading inductor size against switching loss and the >88% efficiency floor. If a specific band must be avoided (e.g. AM broadcast, 530-1710 kHz), say so, since that would rule out several otherwise-suitable 1 MHz-class parts. Note the requirement interacts with Q31. 33. **Silkscreen content?** Default: board name "sbuck-5v3a", revision "A", date, input and output voltage/current markings at each terminal, polarity (+/-) on both connectors, pin-1/polarity marks on every polarised part, and every test point labelled. No logo.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Unattended run: orchestrator answered all 33 requirements open questions as delegate (requirements-answers.md)	Brief DECISION POLICY forbids clarifying questions; no mains/battery/motor/>30V/RF flags apply, so no safety unknown is being guessed
P0	4-layer stackup, 1oz outer / 0.5oz inner	Brief demands an uncut GND plane directly under the switching components plus 2.05 W natural-convection cooling on <=50x40mm; 2L cannot give both the image plane and the bottom thermal copper
P0	Add a 4A slow-blow input fuse (DCR <=30 mOhm) beyond the brief's protection list	IC OCP+hiccup cannot protect against a shorted high-side switch; upstream supply current limit is unstated, so the fuse is what bounds fault energy on-board

Phase	What	Why
P0	Target switching frequency 400-700 kHz	Below the AM broadcast band, keeps 18V-input min-on-time and switching loss comfortable, and allows 20-40% ripple with a stocked inductor
P0	Spawn tier deviation: Fable 5 unavailable (out of credits); judgment roles run on Opus instead	Playbook forbids silent downgrade - Opus is the nearest available tier for architect/placement/reviewer roles; recorded rather than hidden
P1	Input fuse: take the 5A slow-blow (Bel 0685T5000-01, 5032 pcs) over the spec-ideal 4A sibling (174 pcs)	Requirements Q24 binds stock ≥ 2000 pcs and the brief says an out-of-stock part is a failed design; 5A still opens on a bolted fault and costs only 119 mW at the 2.44 A worst case
P1	P3 must pull screw-terminal body height, pin diameter and recommended hole size from the vendor datasheet - LCSC attributes do not carry them	Footprint cannot be built or DFM-checked on missing mechanical data; flagged by the protection scout as its open item
P1	All MLCCs must be X7R (no X5R anywhere); Cout bank moves to 22uF/25V 1210 X7R	Architect computed 83-87 C board surface; X5R is rated 85 C so it is a latent failure, and the X7R part also derates less on DC bias ($\sim 12\%$ vs $\sim 25\%$)
P1	Add a 15V Zener + gate resistor clamp on the reverse-polarity P-FET gate	AO4407A survives 18V static on its +/-25V Vgs, but the architect's 25.4V hot-plug ring on the screw terminals exceeds it; the clamp is the cheap fix
P1	In1 and In2 both solid GND; no power net may use an inner layer	0.5oz inner copper needs 3.0-4.6mm widths for these currents per IPC-2152, which is unroutable - so power stays on 1oz outer and the inners are pure return/thermal
P1	Feedback divider uses 0.1% resistors regardless of IC choice	Reference tolerance alone spans -1.2%/+1.8% across the candidates and already eats 60-95% of the +/-2% budget; FB bias current error is negligible ($\leq 0.1\%$) so the divider tolerance is the only lever left
P1	RC snubber values come from general practice, not a vendor citation - footprint sized generously and left DNP	None of the three candidate vendors publishes a snubber value; the brief wants the footprint anyway, so honest sourcing is to mark it uncited rather than invent a citation
P1	P2 may revisit the 4.0A current-limit-minimum floor - it was a delegate tightening (Q5), not a user requirement	It is the single rule disqualifying LMR33630 (3.85A min, ~ 3.375 A guaranteed Iout = 12.5% over the 3.0A load), and the AP64350 that satisfies it carries a 7-10mA UVLO penalty and external compensation
P2	OVERRIDE of delegate answer Q14: UVLO retargeted from 6.5V/6.0V to 6.2V/5.3V	0.5V hysteresis is less than this board's own 0.49V cable drop at 2.44A (motorboating), and VON=6.5V at the max threshold corner lands ~ 7.0 V so the converter could refuse to start at its minimum rated input; also drops the UVLO divider draw from 81-181mW to 1-2mW
P2	Buck IC: AP64350SP-13 at 500 kHz (RT=200k)	Decided on MAX Rds(on) 75/45 mOhm vs LMR33630's TYPICAL 95/66; TI's own max column gives 1.65W conduction $\rightarrow$ Tj ~ 112 C, over the 105C limit. SY8205 rejected on preliminary datasheet with typ-only Rds

Phase	What	Why
P2	1 oz outer copper; both vendor reference layouts (2 oz) lose	JLC's only 4L/1.6mm 2oz-outer lamination has a 0.4284mm L1-L2 prepreg vs 1080B's 0.2444mm - nearly doubles thermal-via resistance and pushes the image plane 75% further from the hot loop; net +0.8C, so 2oz is worse here
P2	Stackup JLC04161H-1080B; F.Cu + In1 + In2 + B.Cu ALL GND	JLC inner copper is 0.0152mm - thinner than the nominal 0.5oz the rule assumed - so inner power widths would need 3.5-5.3mm; declared explicitly because the pipeline default would otherwise pour In2 as +5V
P2	Single flat root sheet (refdes 1-99, #PWR 100), 33 placed parts	Small board; hierarchy rejected to eliminate the /<sheet>/<LABEL> net-name mismatch class the recipe warns about

Sheet plan

sbuck-5v3a - schematic sheet plan, net contract and refdes map

1. Sheet plan: ONE flat root sheet

```
| Sheet | Blocks ('blocks.md') | Refdes block | '#PWR' base |
|---|---|---|---|
| **root** (no child sheets) | B1-B9, all of them | **1-99** | **100** |
```

Decided: a single root sheet, no hierarchy. 33 placed parts (2 of them DNP) plus 7 test pads and 4 mounting holes. This is one function - a single-rail converter - with one floorplan that runs straight through from the left edge to the right edge, so there is no organising principle for a split to express. The full-run recipe explicitly allows one schematic agent for a board this size.

The decisive argument is the **net-naming risk**, not the part count. KiCad names a net from a hierarchical label as /<sheet>/<LABEL>, and every `constraints.json` consumer at P5-P8 matches on the exact string; a mismatch is silently no-op'd rather than failing. That trap cost the lumina-par run a P4 amendment and forced rf-de-20m into a binding root-label contract for a single crossing net. A flat sheet removes the entire class: local labels come out as /NAME and power symbols come out bare, with no sheet prefix anywhere.

Refdes are unique by construction on one sheet. The blocks below are a *readability* convention for the BOM, not a uniqueness mechanism.

2. Net-naming contract - BINDING ON P4

These are the CANONICAL names. `constraints.json` matches them literally.

```
| Canonical net | Kind | Carries |
|---|---|---|
| '/VIN' | local label | J1.1 -> F1 -> Q1 drain. **2.44 A at 7 V**, raw and unprotected |
| '+VIN' | **power symbol / global label (bare)** | Q1 source -> C4-C9 -> U1 VIN. 2.6 A design, 1.5 A rms in the ceramics |
| '/SW' | local label | U1 SW -> L1. **3.53 A pk, hard-switched, 500 kHz** |
| '+5V' | **power symbol / global label (bare)** | L1 -> C10-C15 -> J2. 3.3 A design ceiling |
| 'GND' | **power symbol (bare)** | F.Cu island + In1 + In2 + B.Cu. 3.3 A
return |
| '/EN' | local label | R2/R3 junction -> U1 EN. TP5 |
| '/FB' | local label | R6/R7 junction -> U1 FB. High-Z, keep away from '/SW' and L1 |
```

```
| '/COMP' | local label | U1 COMP -> R5 -> C2 -> GND |
| '/BST' | local label | C1 -> U1 BST |
| '/RT' | local label | U1 RT -> R4 -> GND |
| '/QGATE' | local label | Q1 gate, R1 to GND, D2 anode |
```

Rules P4 must not break:

1. +VIN, +5V and GND must be **power symbols or global labels** so they come out bare. A local label would produce /+VIN and five `constraints.json` entries would stop matching in silence. 2. Every other net above must be a **local label on the root sheet**, producing the leading /. Do not "tidy" the slash away. 3. ****Do not rename anything to end in _P, _N, _H, _L, DP, DM, + or -.**** `rules_gen.detect_diff_pairs` auto-pairs `high_speed` nets with those suffixes; /SW is declared under `high_speed` and a suffix collision would silently emit differential-pair gap rules and an inner-layer track ban on the most current-carrying node on the board. Current names are safe. 4. `diff_pairs` is an **explicit empty list** in `constraints.json` - there is no differential net on this board and the empty list disables auto-discovery.

3. Refdes map

B1 input entry + reverse polarity (/VIN, /QGATE)

```
| Ref | Part class | Note |
|----|----|----|
| **J1** | 2-pin 5.08 mm THT screw terminal, DB128L-5.08-2P class | **LEFT short edge.** DNP for assembly, hand-soldered (Q25) |
| **F1** | 5 A slow-blow 1206 chip fuse, Bel Fuse C1T class | 63 V DC, 20 mOhm, 5.3 A^2s. >= 20 mm^2 copper per pad, kept off the U1/L1 zone |
| **Q1** | P-channel MOSFET SOIC-8, AO4407A class | **drain to input, source to load.** >= 50 mm^2 of drain+source copper |
| **R1** | 100k 0603 | gate to GND. Also the polarity-reversal turn-off path |
| **D2** | 15 V Zener SOD-123 | anode at gate, cathode at source. Clamps \|Vgs\|
```

B2 input capacitance (+VIN)

```
| Ref | Part class | Note |
|----|----|----|
| **C4** | 100 uF / 35 V hybrid-polymer alu SMD can, KNM2100UF35V149EC0055 class | **ESR 50-300 mOhm is a REQUIREMENT.** Place at the input edge (coolest corner). Not a low-ESR polymer, not a plain electrolytic |
| **C5-C8** | 4x 4.7 uF / 50 V X7R 1206, 1206B475K500NT class | 1.5 A rms shared four ways |
| **C9** | 100 nF / 50 V X7R 0603 | **AT the U1 VIN pin.** Innermost element of the hot loop |
```

B3/B4 converter, compensation, UVLO (+VIN, /SW, /EN, /FB, /COMP, /BST, /RT)

```
| Ref | Part class | Note |
|----|----|----|
| **U1** | AP64350SP-13 class, SO-8-EP | Single GND pin + exposed pad on the same net. No VCC pin |
| **C1** | 100 nF / 50 V X7R 0603 | BST to SW. Required, not optional |
| **R4** | 200k 1% 0603 | RT -> 500 kHz ('RT[kOhm] = 100000/fsw[kHz]') |
| **R5** | Rcomp, 0603 | vendor start 14k. **P4 RE-DERIVES against 5x22 uF** |
| **C2** | Ccomp, 0603 | vendor start 3.3 nF. **P4 re-derives** |
| **C3** | 47 pF COG 0603 | optional feedforward across R6 |
| **R2** | **105k 1% 0603** | +VIN -> EN. UVLO top |
| **R3** | **24.0k 1% 0603** | EN -> GND. UVLO bottom. Target VON 6.2 V / VOFF 5.3 V |
```

B5/B6 output filter and feedback (/SW, +5V, /FB)

Ref	Part class	Note
---	---	---
L1	6.8 uH molded alloy-composite, FAUL1050-6R8MT class	11.5 x 10 mm, 4.1 mm. **>= 125 C rated, DCR <= 25 mOhm at 20 C**
C10-C14	5x 22 uF / 25 V X7R 1210, TCC1210X7R226K250MT class	**4x is the authorised floor, 3x is not** (blocks.md B5)
C15	4.7 uF / 16 V X7R 0805	output HF bypass at the terminal / test point
R6	~116k **0.1%** 0603	FB top. 0.5% is the absolute floor
R7	~22.1k **0.1%** 0603	FB bottom. Ratio 6.249 -> 5.000 V from a 0.8 V ref

B7/B8/B9 indicator, output, snubber

Ref	Part class	Note
---	---	---
D1	green 0805 LED, KT-0805G class	~1 mA
R8	2.2k 0603	LED series
J2	same class as J1	**RIGHT short edge.** DNP for assembly
R9	0603, **DNP**	snubber R. 10-33 Ohm if ever fitted
C16	0603, **DNP**	snubber C. 470 pF - 2.2 nF if ever fitted
TP1-TP7	1.5 mm bare SMD round pads	'+VIN', '/SW', '+5V', '/FB', '/EN', 'GND', and a scope-ground pad ~5 mm from TP3
H1-H4	3.2 mm NPTH	corners, inset 3.5 mm, 6.5 mm keepout, isolated from GND

Total: 33 placed components (2 DNP), 7 test pads, 4 holes.

4. Floorplan for P6 - straight-through, left to right

The board is 50 mm wide (x) by 40 mm tall (y); the **short edges are left and right**, which is where the two connectors go (Q10).

Zone	x	Contents
---	---	---
input	0 - 13	J1, F1, Q1, D2, R1, C4. The coolest corner - C4's life is set by local board temperature
converter	13 - 32	C5-C9, U1 (thermal island, centred near x=21 y=20), C1, R2-R5, C2, C3, /SW pour, R9/C16, L1
output	32 - 50	C10-C15, R6/R7, D1/R8, TP1-TP7, J2

Non-negotiables that no annealer can discover:

1. **Hot loop first.** C5-C9 -> U1 VIN -> U1 PGND -> C5-C9 is the shortest loop on the board, all on F.Cu, with C9 physically closest to the pin and In1 solid GND directly beneath as the image plane.
2. **'/SW' is a short wide pour:** >= 2.5 mm wide, <= 8 mm long, <= 40 mm², F.Cu only, never inner, never near a board edge or connector. The TP2 tap is a stub that counts against the 40 mm².
3. **R6/R7 and the '/FB' trace hug the FB pin and stay away from '/SW' and L1** - an explicit rule in all three candidate vendor layouts.
4. **R9/C16 sit directly across the SW pour to PGND** with a low-inductance path even while unpopulated. Place by explicit place_edit.
5. **C4 and F1 stay >= 10-12 mm from U1 and L1** (declared as separation entries): C4 because its life halves every 10 C, F1 because it is a deliberate 0.15 W heat source.
6. **16 thermal vias in U1's pad** at 0.3 mm / 1.0 mm pitch, plus 8-12 stitching vias in the surrounding F.Cu island. check_thermal will not catch their absence (see power_tree.md s4.3) - this is a review gate.
7. **No layer transitions on '/VIN', '+VIN', '/SW' or '+5V'.** A transition needs 6-8 vias at 0.5 A/via. GND is plane-fed through the stitching field.

Parts to place by explicit `place_edit` and `lock` before the annealer runs: U1, L1, C5-C9, C1, R9, C16, J1, J2, H1-H4.

5. P4 notes (schematic phase)

1. **Re-derive the compensation.** The vendor's Rcomp/Ccomp are quoted for 2x 22 uF; the bank is 5x 22 uF. Target fc 25-50 kHz, phase margin ≥ 45 deg, using the AP64350 datasheet's Eqs 12-20. Record the numbers on the schematic. If the loop cannot close there, drop to 4x 22 uF and record it. 2. **R6/R7 must be 0.1% parts in the BOM**, not 1% with a 0.1% note. The reference tolerance alone spends 60% of the $\pm 2\%$ window. 3. **UVLO is 6.2 V / 5.3 V, not the delegate's 6.5 / 6.0.** Use the AP64350 divider equations (they include the 4.114 uA and 5.5 uA internal pull-ups - a naive divider calculation is wrong) and verify against the P8 sim bench. 4. **No gate RC on Q1 and no gate capacitor.** The body diode already bypasses the channel during inrush. This is a decision, not an omission - do not let ERC or a reviewer add one. 5. **The snubber R9/C16 must exist in the netlist** as DNP so their pads, clearance and routing are accounted for at P6/P7. Populating later is then a BOM change, not a respin. 6. **Test pads are real nets.** TP7 is a second GND pad placed for a scope spring ground, ~5 mm from TP3, not a duplicate of TP6. 7. **Silkscreen (Q33):** board name + rev A + date; "VIN 7-18V" and "VOUT 5V 3A" with polarity marks at both terminals - this is the *only* mitigation for a swapped-connector event, which is not survivable by design; pin-1 marks on every polarised part; every test point labelled; no logo. 8. **J1 and J2 are DNP for assembly** and must appear as such in the BOM/CPL.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Pending — produced at P4.

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake digest - sbuck-5v3a

- Function: open-frame synchronous buck module. 7-18 V DC in (12 V nom) ->
5.0 V
+/-2% at 3 A continuous (15 W). No MCU, no digital interface.
- Artifacts: 'brief/brief.md', 'requirements.md' (analyst, 9/9 sections,
```

check_requirements pass, 0 violations), 'requirements-answers.md' (delegate).

- 33 open questions answered as DELEGATE per the brief's DECISION POLICY. No safety unknown was guessed: no mains, no battery, no motors, nothing >30 V, no RF transmit. 3.0 A sits ON the >3 A threshold - treated as high-current for copper/connector sizing, no certification path triggered.
- Binding constraints: 50 x 40 mm HARD (binds at P5 board_init), 4x M3 3.2 mm NPTH, 50 C free-air ambient with no airflow, every part LCSC-stocked.
- Derived arithmetic handed forward so P1/P2 do not re-derive: $D = 0.71/0.42/0.28$

at Vin 7/12/18 V; Iin = 2.44/1.42/0.95 A; input-cap RMS current peaks at 1.50 A at Vin = 10 V (INSIDE the range, not at 12 V nominal); loss budget 2.05 W at the 88% efficiency floor.

- Riskiest item: thermal. 2.05 W by natural convection at 50 C ambient on a <=50x40 mm board with no heatsink is the binding case, and the brief forbids assuming the exposed pad alone closes it.
- Decisions recorded: 4-layer 1oz/0.5oz; 4 A slow-blow input fuse added beyond the brief's protection list; fsw target band 400-700 kHz; Opus substituted for Fable across judgment roles (out of credits) - recorded, not silent.
- Second-riskiest: input and output connectors are physically identical, so a user swap is a real failure mode. Resolution is silkscreen only (Q30).

P1

P1 Research digest - sbuck-5v3a (detail: log/P1-findings-full.md)

- Roster: 3 scouts + power architect + reference-design, plus 1 re-scout to correct magnetics against the architect's budget. All fragments in research/.
- TPS54360 and MP1584EN are ASYNCHRONOUS per their own datasheets - disqualified.
- Every ferrite-wound inductor fails Isat/Itemp derated to 50 C; metal-composite only.
- First inductor sweep missed the DCR ceiling 2x by only searching <=6x6 mm.

Re-search:

DCR <= 25 mOhm reachable (FAUL1050-6R8MT, 18.5 mOhm cold / 24.05 mOhm hot).

- AP64350 UVLO divider draws 7-10 mA (~120 mW, 6% of budget); LMR33630 does it at 23 uA.
- Loss budget IS the thermal budget: 19 C/W per watt, worst Tj 99.7 C vs 105 C limit.
- Exposed pad alone = 138 C/W -> Tj 188 C. Package rule: pad + thetaJA <= 45 C/W.
- Inrush benign (body diode forward - a gate RC cannot limit it); real hazards are the 25.4 V hot-plug ring (15 V Vgs Zener) and bulk ESR as damping (50-300 mOhm, no polymer).
- Board surface 83-87 C forces X7R everywhere; X5R is a latent failure.
- 4 research conflicts resolved explicitly (DCR, bulk ESR, dielectric, Vgs clamp).
- Left for P2: IC choice (4.0 A floor is revisitable - it was a delegate tightening), fsw coupled to inductor stock, and vendor 2 oz layouts vs "2 oz does not help".

P2 (no digest recorded)

Gate attempt history

(no entries)

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	not reached		
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
state.json	11,967 B	
brief/brief.md	5,387 B	
requirements.md	20,496 B	
architecture/blocks.md	12,147 B	
architecture/constraints.json	16,342 B	
architecture/decisions.md	15,216 B	
architecture/power_tree.md	12,708 B	
architecture/sheets.md	9,927 B	
architecture/stackup.md	7,559 B	